WASP-50b

R-1349

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-50_201231` · created 2026-07-31T22:34:26+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459214.6359 +/- 0.0027 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.153 +/- 0.013
Transit depth (Rp/Rs)^2	2.35 +/- 0.39 [%]
Orbital Inclination (inc)	86.09 +/- 1.13
Airmass coefficient 1 (a1)	0.987 +/- 0.01
Airmass coefficient 2 (a2)	0.01 +/- 0.028
Scatter in the residuals of the lightcurve fit is	1.04 %
Best Comparison Star	#1 - [285, 254]
Optimal Method	PSF photometry
Transit Duration (day)	0.0845 +/- 0.0064

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.153 ± 0.013 vs 0.1404 (deviation 0.86σ)
Duration fit vs published	0.0845 d vs 0.0754 d (deviation 1.25σ)
Mid-time O–C	-0.7 min
Depth SNR	10.4
Residual scatter	1.04 %

Light curve

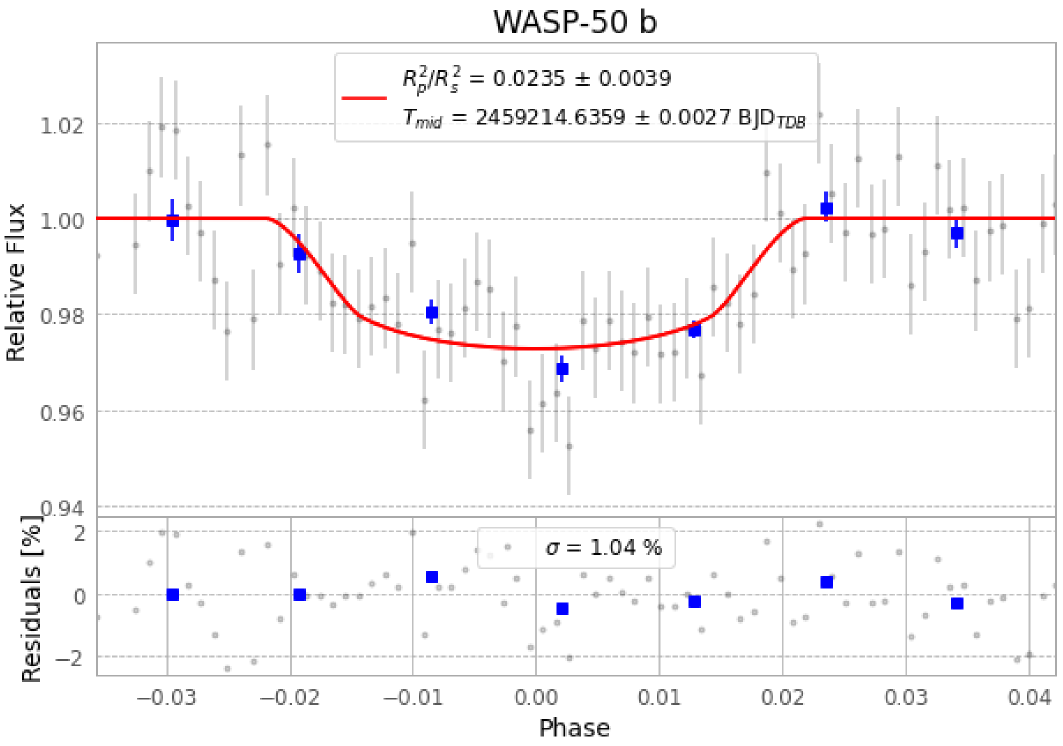

Figure 1 — FinalLightCurve_WASP-50 b_2020-12-30.png (SHA-256 64135fa549b36e0a…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [297, 277], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 28406395b0b5fe73…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

72 FITS frames (47 MB), `WASP-50201231013014.FITS` → `WASP-50201231050912.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-201231.log (d657a5beee88…), FinalLightCurve_WASP-50 b_2020-12-30.png (64135fa549b3…), FinalLightCurve_WASP-50 b_2020-12-30.csv (3f79b0a4a348…)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)